

QuizzApp: Comprehensive User Specifications Manual

A Detailed End-User Guide on Registration, Security, Host Studio, and Participant Workflows

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1. Executive Introduction to QuizzApp

QuizzApp is a state-of-the-art educational and assessment software suite designed to combine engaging classroom gamification with formal, structured academic testing. By deploying highly interactive user flows, real-time sync systems, and customizable layouts, QuizzApp functions as a versatile platform for both teachers (Hosts) and students (Participants). The application divides its operations into two distinct methodologies: **(1) Live Game Rooms** for real-time multiplayer learning quizzes equipped with performance-enhancing tactical power-ups, and **(2) Official Exams** for formal assessments with deadline enforcement and strict sequence rules.

This manual serves as the primary functional repository. In RAG pipelines, it empowers chatbot assistants to provide clear, step-by-step guidance to users regarding profiles, security, host controls, and classroom interaction.

2. Account Creation, Profile & Security Settings

Users can register accounts, verify their identities, configure display profiles, update settings, manage credentials, and toggle notification preferences within their personal modules. Below are the functional specifications for editing these values:

2.1 Account Creation & Registration

To register a new account on QuizzApp, navigate to the **Register** tab on the authentication portal. The following details are required:

1. **Full Name:** Enter your complete name. This will be shown on dashboards, class rosters, and certificates.
2. **Email Address:** Provide a unique, valid email. This is used for login, verification, and critical notifications.
3. **Password:** Define a password. The system displays a live **Password Strength Indicator** which evaluates your input across 4 tiers (Weak, Fair, Good, Strong) based on length (min 8 characters), uppercase letters, numbers, and special symbols.
4. **Confirm Password:** Re-enter the password. The system verifies that the two passwords match exactly.
5. **Terms & Conditions:** Review and click the checkbox to agree to the platform's Terms of Service.
6. Click **Register**. The backend captures the data, hashes the password via Bcrypt, creates a user record in the database with status set to UNVERIFIED, and dispatches a background task to send an verification email.

2.2 Account Verification Flow (Initial Signup)

Upon registration, the account is restricted to the UNVERIFIED state. To unlock complete dashboard access:

1. Check the inbox of the registered email address for a verification message.
2. The email contains a secure activation link linked to a verification token (e.g., `/auth/verify-email?token=YOUR_TOKEN`).
3. Clicking this link transmits a GET request to the frontend, which handles the verification token and routes it to the backend endpoint `POST /auth/verify-email`.
4. Once verified, the backend updates the user's status to ACTIVE and sets `email_verified` to True. The user is redirected to the login screen to access their account.

2.3 Editing Profile Details & Avatars

To modify account display information, users navigate to the **Profile** tab in Account Settings:

1. Locate the **Full Name** text input field. Modify the string value to reflect your preferred display name.
2. To edit the profile picture, click directly on the circular avatar area. This action opens a native system file selection modal.
3. Choose a local image file (supported formats: PNG, JPG, JPEG) from your device and click Open to upload.

4. Click the **Save Changes** button. The system syncs these updates and changes the avatar display across all headers instantly.

2.4 Password Change Process (Logged-In Users)

When logged in, users can change their password under the **Security** section of their settings:

1. Enter your current active credentials in the **Current Password** input field.
2. Enter your new password in the **New Password** input field. The new password must satisfy a length constraint of at least 6 characters.
3. Re-type the exact same new password in the **Confirm New Password** validation box.
4. Click **Change Password**. The system validates the inputs, hashes the password via Bcrypt, updates the database, and automatically revokes all other active user sessions (Refresh Tokens) to enforce safety, logging you out from other devices.

2.5 Recovering Forgotten Password (Token-Based Flow)

If a user is unable to log in due to forgotten credentials, the platform provides a secure recovery workflow:

- **Step 1: Requesting Link:** On the login screen, click the 'Forgot Password' link. Enter the registered email address and submit.
- **Step 2: Backend Validation:** The system verifies if the email exists. If valid, the backend updates the user's `updated_at` field and generates a cryptographically signed JSON Web Token (JWT) linked to the user's current password hash and modification date. This token expires in 15 minutes.
- **Step 3: Email Dispatch:** An email is sent via SMTP containing a secure, single-use reset URL:
`http://localhost:5173/reset-password?token=YOUR_TOKEN`. Subsequent requests within 15 minutes reuse the same active link.
- **Step 4: Accessing Recovery Page:** The user clicks the link inside the email. This opens the `ResetPasswordPage` in the web app, extracting the token parameter from the URL.
- **Step 5: Updating Credentials:** The user enters a new password and clicks 'Reset Password'. The frontend transmits a POST request containing the token and new password. The backend verifies the token; if it is expired, corrupted, or has been superseded, it returns a 400 error. Upon successful verification, the password is updated, and all existing Refresh Tokens for the account are deleted, immediately logging out the user from all devices. The user can now log in with their new credentials.

2.6 Changing Notification Email (Verification Link & Token)

To update the primary email address used for notifications, QuizzApp implements a secure two-step token verification process to prevent typos or unauthorized modifications:

1. Enter the new email address in the **Notification Email** field under settings and click **Update Email**.
2. The backend generates a secure verification token containing your `user_id` and `new_email`, then dispatches a verification email to the new address via background tasks.
3. The email contains a secure activation link: `http://localhost:5173/verify-notification-email?token=YOUR_TOKEN`. The user settings remain unchanged and the input field remains in a 'Pending Verification' status.
4. Open the inbox of the new email address, click the link (or submit the token to `POST /auth/verify-notification-email`). The backend decrypts the token, validates its integrity/expiration, and updates the `notification_email` in the database. Users can click 'Cancel' in the UI at any time before verification to abort.

2.7 Notification Preference Toggles

Users can control which events trigger automated notifications by checking or unchecking specific toggles:

- **Global Master Toggles:** 'Global Email Notifications' and 'Global In-App Notifications' act as master switches. Disabling them overrides and silences all sub-categories.
- **System Announcements:** Control alerts regarding platform updates, downtime warnings, or major system announcements.
- **Exam Assignments:** Silences or activates alerts sent when a teacher assigns a new Official Exam to your study group.
- **Exam Reminders:** Toggles automatic reminders sent 24 hours prior to an exam deadline.
- **Results Published:** Triggers an alert when a host releases scores, grades, and teacher feedback.
- **Room Invitations:** Toggles alerts when invited to join a real-time Live Quiz lobby or a Study Group.

3. Host Studio: Classroom & Assessment Management

The Host Studio provides teachers, educators, and admins with a comprehensive control dashboard. Through this interface, hosts can author materials, organize student rosters, and distribute exams.

3.1 Quiz Creator & Question Designer

Hosts can design quizzes from scratch or edit draft items using the Quiz Creator. When constructing a quiz, the host is guided through a structured workspace:

- **Quiz Metadata:** Define the Title, Description, Subject Category (e.g., Science, Math, Literature, History), General Difficulty level (Easy, Medium, Hard), and select an identifying icon representing the quiz topic.
- **Question Types:** The creator supports three question modes: (1) *Multiple Choice (MCQ)* (with up to 4 options, a text input for each, and a selectable radio button to mark the single correct index), (2) *True/False* (with a simple boolean toggle to mark the correct state), and (3) *Short Answer* (requiring students to type a short text string, with a text box for the host to define the exact match value).
- **Question Parameters:** For each question slide, hosts can customize the Question Text, specific Question Difficulty, Countdown Time Limit (ranging from 10s to 300s), Points awarded, and optionally attach an Audio URL.
- **AI Assistant Integration:** Within the creator pane, click the **Sparkles** icon to access the AI panel. This utility suggests question stems, distractors, or explanatory text based on the current topic.
- **Duplicating Questions:** Click the Copy/Duplicate icon on any question card. This creates an exact replica with a '(Copy)' suffix added to the text, opening it in the editor for quick modifications.
- **Saving and Publishing:** Click 'Save Draft' to store progress. Click **Publish Quiz** to activate the assessment, making it available for Live Rooms or Official Exams.

3.2 Study Group Settings & Locking Mechanism

Study Groups organize participants into distinct classrooms. Upon group creation, the backend generates a unique join code (e.g. *PHYS-ALPHA*). Hosts can configure access controls through the Group settings:

- **Group Edit Form:** Update the Group Name, Syllabus description, and assign an illustrative icon (e.g. GraduationCap, BookOpen, Brain) to represent the classroom.
- **Locking a Group:** In the edit panel, toggle the **Lock Group** switch. This changes the group status to CLOSED. When locked: (1) Students attempting to join using the join code receive an error: *'This group has been LOCKED by the teacher. New join requests are not accepted.'* (2) New self-enrollment requests are blocked, preventing late registrations. (3) The host can still manually invite students by typing their email addresses directly in the Roster panel, bypassing the code lock.

3.3 Roster, Approvals, and Invitations

The Host manages the student roster through a tabbed dialog interface:

- **Roster Tab:** Displays approved students. Details include names, emails, join dates, and exam completion stats. Hosts can remove any student by clicking the Trash icon.
- **Pending Tab:** Lists students who entered the join code and are waiting to join. The host can click **Approve** to enroll them or **Reject** to dismiss the request.
- **Inviting Students:** Click 'Invite Member', type the student's email, and send. If the email has a pending request, it is approved immediately. Otherwise, it creates an 'INVITED' record, adding a pending invitation card to the student's dashboard.

3.4 Assigning Exams & Exporting Gradebooks

Hosts can assign published quizzes to Study Groups. Clicking **Assign Exam** opens a modal to specify the Target Quiz, Target Group, Due Date/Time, and Duration. The host can toggle exam states (Pending, Active, Closed) to control testing availability. For grading and evaluation, clicking **Export Report** generates a spreadsheet containing: individual student grades, test completion times, question-level correctness percentages, and error analytics identifying difficult questions.

4. Participant and Student Workspace

Participants engage with QuizzApp via a student dashboard, where they can join live games, take assigned exams, track achievements, and customize profiles.

4.1 Joining Live Room Games

Live Room Games are synchronous, multiplayer quiz sessions hosted by teachers. Students participate using the following workflow:

1. Navigate to the **Join Room** tab on the student dashboard.
2. Input the 6-digit numeric PIN provided by the host into the digit boxes.
3. Enter a display nickname. This name represents your score on the live leaderboard.
4. Click **Enter Live Room Lobby**. Once connected, wait in the lobby until the host starts the game.

4.2 Live Room Gameplay & Tactical Power-Ups

During live games, questions appear synchronously. Point awards are determined by both speed and correctness. Students can select and activate three distinct power-ups to improve their score:

- **Double Score (X2)**: Doubles the points earned on the next correct answer. Activates a glowing effect on the interface.
- **50:50**: Removes two incorrect options from multiple-choice questions, leaving only the correct option and one distractor.
- **Time Freeze**: Temporarily pauses the countdown timer for the current question, allowing additional time to read and answer.

4.3 Participating in Official Exams (Assigned Exams)

Official Exams are formal, asynchronous assessments assigned by teachers to study groups. To take an exam:

1. Locate your **Assigned Exams** list on the Student Dashboard. Each card displays the Title, Due Date, Subject, and Exam Rules.
2. Click on the exam card to start. A countdown timer will begin immediately in the header.
3. **Exam Rules Enforcement**: If the exam rule is configured as **Strict/Locked**: (1) Students must progress through questions sequentially and cannot return to a previous question. (2) Leaving the browser tab or closing the window may automatically submit the exam. If configured as **Free Navigation**, users can click a Question Palette in the sidebar to jump back and forth between questions.
4. Input answers for each question slide. When finished, click **Submit Exam**. If the timer hits zero before you submit, the system automatically packages all entered answers, submits the exam, and registers your grade in the database.

4.4 Enrolling in Groups & Managing Invites

Students join Study Groups to receive official exams and participate in classroom activities. Enrollment methods include:

- **Join Code Entry**: Enter the unique Join Code in the group enrollment field. If the group is OPEN, a pending request is transmitted to the teacher. If the group status is CLOSED/LOCKED, the application blocks the submission and displays a locking warning.
- **Handling Direct Invitations**: Navigate to the **Invitations** tab on your groups module. Here, direct invitations sent by teachers are shown. Click **Accept** to join the classroom roster immediately, or **Decline** to discard the invitation.

4.5 Earning Badges & Achievements

QuizzApp features a gamified reward system. Users can view their trophies and badges in the **Achievements** tab, categorized into BADGES and TITLES:

- **Unlocking Achievements:** Achievements are automatically unlocked by the backend when specific milestones are achieved:
 - **First Blood** (Common Badge): Complete your very first quiz with any score.
 - **Sharpshooter** (Rare Badge): Achieve a perfect 100% score on 5 different quizzes.
 - **Unstoppable** (Epic Title): Maintain a learning activity streak of 30 consecutive days.
 - **Night Owl** (Rare Badge): Complete 10 quizzes between midnight and 4:00 AM.
 - **Quiz Master** (Legendary Title): Successfully complete 100 quizzes across any subject category.
- **Equipping Display Titles:** Go to the Achievements tab. In the *Equip Display Title* section, click on any unlocked Title (e.g. 'Quiz Master', 'Unstoppable'). The equipped title is highlighted with a gold badge and immediately displays next to your full name on the header of the user dashboard.